
APPLIED STATISTICS IN EDUCATION AND HUMAN SERVICES I EDUC 641 FALL 2025

Course Information

Course Title	: Applied Statistics in Education and Human Services I
Course Number	: EDUC 641
CRN Number	: 11867
Number of Credit Hours	: 3
Class Date/Time	: Tuesday & Thursday, 4:00 pm – 5:20 pm
Class Location	: 142 HEDCO Education Building (Click for map)
Instructor	: Dr. Cengiz Zopluoglu (Click for pronunciation)
E-mail	: cengiz@uoregon.edu
Office Hours	: Thursday 2 pm – 3:30 pm (Zoom) or by appointment https://uoregon.zoom.us/j/6883665008?pwd=S1BkMTZhTk4weWZVSjUvQ2FVeGowQT09 Meeting ID: 688 366 5008 Passcode: 768421
Teaching Assistants	: Brittany Spinner (bspinner@uoregon.edu) Nafisul Huq (nhuq@uoregon.edu)
Labs	: Wednesday, 4:00 pm – 4:50 pm, 115 Lokey Ed. Bld. Thursday, 5:30 pm – 6:20 pm, 142 HEDCO Ed. Bld.

Course Modality

This is an in-person course: that means that, unlike asynchronous online/ASync WEB courses, we will meet during scheduled class meeting times in HEDCO 142. I will accommodate absences as described in the Absence policy below. If you need accommodation related to a disability, you can request those by working with the Accessible Education Center (<https://aec.uoregon.edu>).

Course Description

This course provides a first introduction to quantitative data analysis in education and the social sciences. It is part of a three-course sequence intended to provide a toolkit of statistical concepts, methods and their implementation to producers of applied research in education and other social sciences. The course is organized around the principle that research design depends in part on researchers' substantive questions and their quantitative data available to answer this question. In this introductory course, we will focus on describing categorical and continuous data and quantifying the relationship between categorical and continuous data. Students will form a solid foundation for frequentist, inferential statistics (and some of critiques of this approach). The course seeks to blend a conceptual, mathematical and applied understanding of basic statistical concepts. At the core of our pedagogical approach is the belief that students learn statistical analysis by **doing** statistical analysis. This course (or substitute) is a pre-requisite for EDUC 643.

Learning Objectives

By the end of this term, it is expected that students will be able to:

1. Articulate the principles of responsible and ethical quantitative research in education and human services
2. Describe, summarize and visualize quantitative data that are categorical and continuous
3. Describe, summarize and visualize the relationships between quantitative data that are categorical and continuous
4. Conduct a frequentist null-hypothesis significance test of the relationships between simple categorical and continuous data
5. Describe strategies to improve the replicability and generalizability of quantitative research

6. Use an open-source, object-oriented statistical programming language (in this case R) to conduct all such analyses

Course Prerequisites

There is no prerequisite for this course beyond a high school Algebra I course.

Textbook

We will primarily be referring to chapters in [Learning Statistics with R \(LSWR\)](#) by Danielle Navarro. This textbook is available for free online. You may choose to purchase a paper copy if you wish, but it is not required. There are additional readings assignments on the troubling historical origins of modern statistics that appear in the detailed schedule. See detailed schedule.

Other text resources

There are literally dozens of high-quality introductory statistics textbooks. We have particularly found the following to be helpful:

Howell, D.C. (2013). *Statistical Methods for Psychology*. Cengage Learning: Belmont, CA. [A balanced book in terms of technical details and conceptual discussion. A lot of examples and discussions with a context in psychology and social science.]

Fox, J. (2015) *Applied Regression Analysis and Generalized Linear Models*. 3rd Ed. SAGE Publications: Thousand Oaks, CA. [Focused only on regression and a lot more detail in regression diagnostics, remedial measures, etc. compared to Howell]

Darlington, R.B. & Hayes, A.F. (2016) *Regression Analysis and Linear Models*. Guilford Press: New York, NY. [Strong conceptual discussion and many examples]

Gelman, A., Hill, J. & Vehtari, A. (2020). *Regression and Other Stories*. Cambridge University Press: Cambridge, UK. [Focused on applied problems of data analysis, estimation, prediction and causal inference. Computation in R with code available online. Note: this book primarily takes a Bayesian approach to inference and so will be more appropriate as you advance in your learning.]

Statistical Computing

Students must have the latest version of R, which can be downloaded [here](#). It is strongly recommended that students also download the RStudio GUI, available [here](#). Both softwares are free. The R bootcamp will provide a tutorial on R/RStudio installation. While we will teach you how to effectively use R and RStudio to conduct analyses, one of the key skills required to use R is the ability to find answers on your own. Many common questions or problems are available on blogs, have been asked and answered in

discussion forums already, or can be problem-solved using the assistance of Artificial Intelligence tools. Finding and deciphering those answers is an important skill you should seek to hone. *You will never remember all of the programming commands!*

Here are some sites where you can find the answers to many R questions and learn new tricks:

- [YaRrr, the Pirate's Guide to R](#) by Nathaniel Phillips
- [R for Data Science](#) by Hadley Wickham
- [R Cookbook](#) by Winston Chang
- [Cheat Sheets](#)
- [Quick-R](#)
- [StackOverflow](#)
- [Stack Exchange](#)

Course Website

Canvas is an essential part of our course and the place for course information and engagement outside class time. Log into canvas.uoregon.edu using your DuckID to access our class. If you have questions about accessing and using Canvas, visit the [Canvas support page](#). Canvas and Technology Support also is available by phone or live chat:

541-346-4357 | livehelp.uoregon.edu

I will use Canvas as the main website for organizing the course content. Everything related to the course content is also hosted in a GitHub repo you can access (https://github.com/czopluoglu/educ641_f25) and a public course website (https://czopluoglu.github.io/educ641_f25). The Canvas course website contains links for lecture notes, assignments, readings, datasets, etc., and the homework assignments will be submitted via Canvas.

Course Structure

The course will consist of a combination of in-person lectures, required readings, optional lab activities, homework assignments, quizzes, and a final project writing.

In-person lectures

The lectures and the course content are primarily provided in class during the scheduled time and location. You are encouraged to overview the slides, study the material provided, and complete the readings before the class and bring your questions to have a more active discussion about the content. The lecture material and course content will be posted on Friday (or earlier) for the upcoming week. Depending on the technology available in class, I will try to record the in-person lectures and post the link on Canvas.

Readings

These readings are primarily chapters from books available online at no cost to you. They contain relevant technical material about the content covered and will supplement the content provided in lecture notes.

Homeworks

There will be four homework assignments. The goal of the assignments is to practice the concepts and vocabulary we have been modeling in class and implement some of the techniques we have learned. Each assignment has an associated data source, short codebook and detailed instructions for the required data and analytic tasks. You can work with classmates on these activities and discuss your work with others; however, students will upload their completed assignments individually to Canvas. Each student must turn in the completed lab work electronically on Canvas as a Word or Pdf file with your last name as the beginning of the file name.

Quizzes

There will be four take-home quizzes that are designed to test your knowledge of theoretical principles underlying the statistics we are studying for the week. Quizzes will be open book, notes and computer, and must be completed all by yourself without using any AI tool.

Final Project

There is no final exam in this class. Instead, you will submit a sample writing as a final project. For the final project, you should find a dataset, including some variables you are interested in. This writing should be no longer than three pages (single-spaced), excluding the tables and figures. This writing should include the following components:

- **Introduction:** This introduction will briefly provide a context and background and describe variables in your study. The introduction should also include a written research question that can be answered using one of the methods discussed in the class. You should have at least two and no more than three research questions.
- **Method:** This section should describe the sample and dataset by providing the descriptive statistics of the sample characteristics and independent and dependent variables in your research questions. This section should also mention the type of analysis you plan to use for answering your research question.
- **Results:** This section should provide a written summary of your results and interpretation of your statistical analysis, supplemented by APA style formatted tables and figures. If you can't fit the table and figures into the 3-page limit, you can provide them in the Appendix.
- **Appendix:** This section should include all R syntax/output as an appendix for reference (this is not included in the 3-page limit).

Below is a few links you may use to explore publicly available datasets:

- [Inter-university Consortium for Political and Social Research](#)
- [Google Dataset Search](#)
- [Discover Mendeley Data](#)
- [Miscellaneous Datasets](#)
- [PISA](#)
- [TIMSS and PIRLS](#)

You are allowed to discuss with your advisor or other faculty and students to pick an interesting research question and find a relevant dataset. On the other hand, doing analysis, interpreting the output, and summarizing your findings must be all your independent work. You are not allowed to discuss your writing sample with others until it is graded. A detailed rubric is provided below for the final project writing.

FINAL WRITING RUBRIC		Points
Brief Introduction (Recommended length: 400 words)		15
1. Is there brief background information to provide some context for the research question?		5
Tell us why your research question is essential and why it matters.		
2. Is there a problem statement that is clear and concise, includes all variables to be considered, and can be responded through the collection and analysis of numerical data?		5
Please tell us a research question you can respond to using available quantitative data. As most methods we learn in this class can only handle one independent variable and one dependent variable, your research question should include only one dependent and one independent variable. Don't ask a research question that requires more complicated analyses than what is covered in this class (t-tests, ANOVA, chi-square, correlation, simple linear regression)		
3. Are the dependent and independent variables accurately identified? Are the constructs these variables represent clearly defined and described?		5
Method (Recommended length: 500 Words)		35
4. Are the characteristics of the sample dataset described?		10
Show us a table or some text providing a brief demographic description of your participants.		
5. Are the descriptive statistics for the variables of interest adequately described?		10
Show us a table and/or a plot, or some text providing descriptive statistics for all the variables (dependent and independent variables). Describe their distribution.		
6. Is the method being used explicitly spelled out?		5
Please tell us how you will respond to your research question. Are you going to use a dependent samples t-test, independent samples t-test, simple linear regression, chi-square test, or just a correlational analysis?		
7. Are the types of variables aligned with the method being used?		10
See Appendix A below for a summary and more information.		
Results (Recommended length: 600 words)		30
8. Are the findings adequately summarized in an APA-style write-up?		30
Provide a paragraph to summarize the results of your analysis. Make sure you follow the APA style when reporting the results of statistical analysis. Make sure you report them accurately. I will cross-check them with the software output provided as an Appendix.		

General Guidelines	20
9. Is the final project paper less than 3-pages (or 1500 words)? The page limit is stringent. The page limit excludes tables, figures, and any software code/output provided as an appendix. You can insert the tables and figures in the text or provide them separately at the end.	5
10. Is there an Appendix including all R syntax/output for reference? Make sure you include all R syntax/output for reference at the end as an Appendix. It is sometimes helpful if I don't understand things in the write-up. I can go and look from where you get the numbers.	5
11. Are all tables and figures follow APA guidelines? Make sure you create tables and figures following APA style if you include them in your final project paper. Please do not copy/paste software output in your 3-page writing. Everything you want to present should be in properly formatted tables and figures.	5
12. Are the references included? References are excluded from the 3-page limit.	5

Grading

The grades will be combined using a weighted average grading scheme with the corresponding weights given below:

- 4 Homework assignments 60% (15 % each)
- 4 Quizzes 10% (2.5% each)
- Final Project (Report and GitHub Repo) 30 %

Grades will be assigned based on the percent of the total possible points accumulated on the homework assignments, quizzes, and final project. Final letter grades will be assigned according to the following scheme:

A+	97% - 100%	B	83% - 86.99%	C-	70% - 72.99%
A	93% - 96.99%	B-	80% - 82.99%	D+	67% - 69.99%
A-	90% - 92.99%	C+	77% - 79.99%	D	63% - 66.99%
B+	87% - 89.99%	C	73% - 76.99%	D-	60% - 62.99%
				F	< 59.9%

If this class is taken P/NP, 80% or higher is required. Late assignments without notice to the instructor will be penalized such that 20% of total points are deducted for each day it is late.

Course Incomplete Policy

Students are expected to be familiar with university policy regarding grades of “incomplete” and the timeline for completion. For details on the policy and procedures regarding incompletes, Please see: <https://registrar.uoregon.edu/current-students/incomplete-policy>

Office Hours

Office hours are on Thursdays from 2 pm to 3:30 pm and will be held via Zoom ([Link](#)). If you prefer an in-person consultation during office hours, please send me an email. If these times do not work for you, please send me an email to schedule another day and time.

Communication

Our class will communicate through our Canvas site. Announcements and emails are archived there and automatically forwarded to your UO email, and can even reach you by text. Check and adjust your settings under Account > Notifications. I do so through Canvas messages when I need to get in touch with individual students. When giving feedback on assignments, both I and GE will do so in Canvas, and the turnaround time for feedback is generally one week.

If you have a question about course content, assignment, reading, or other class components, please post your question on the Discussions section which I respond to daily, and where your peers can also pose questions and share answers. If your question is about a technical challenge with Canvas, Zoom, or another technology, please contact the [UO Service Portal](#). If your question, concern, or excitement is about something personal, is time sensitive, or is something else that doesn't feel like it fits above, please do reach out to me by email or by attending office hours! If you contact me with a question, I will try to respond within one business day.

Absences

This is a face-to-face course. While there is no automatic grade reduction or penalty for missing classes, attendance is essential because we will develop our knowledge through in-class lectures and discussions. This course has a “reason-neutral” policy for absences. You do not have to notify me or explain why you miss classes. If students need to be absent from a class due to illness, I will try to provide a link for the recordings of the lectures so that students can catch up. Students are encouraged to watch these recordings and do their study of class material. Students should contact the instructor for any questions related to the material from lectures they miss.

Student Workload and Time Use

Educational Activity	Hours student engaged	Explanatory Comments
Attendance	24	18 meetings, at 80 minutes per meeting
Class reading and prep	28	Includes reading and review of class material
Homework Assignments	40	4 assignments, 10 hours per assignment
Quizzes	3	5 quizzes, 35 minutes each
Final Project	25	Final project
Total Hours	120	

Artificial Intelligence Use

Students can use GenAI tools in this class to help with course work and assignments unless noted otherwise for specific assignments. Helpful uses include brainstorming ideas, creating outlines, editing, and so forth. However, if you use a GenAI tool, you need to document your use, including the tool you use and when, where, and how in your work process you used it (for example: “I used ChatGPT to generate an outline for my paper, which I then revised before writing my first draft” or “I used slidesAI.io to create the slidedeck style for my presentation.” etc.). In certain cases, as part of your documentation, I may ask you to submit any GenAI results you obtained, so you need to keep GenAI-created drafts and logs of your interactions with GenAI tools; failure to provide such documentation may result in a grade reduction in certain instances. I will

provide helpful resources for how best to use GenAI to support your learning process and work.

Along with documentation of your GenAI use, you are also required to cite GenAI if you use any GenAI-created content in your work submissions, for example text or images or graphics generated by GenAI tools. Please use standard APA or MLA citation guidelines for GenAI, as indicated here: [MLA Style Center](#) and [APA Style Blog](#).

Grievance Policy

A student or group of students of the College of Education may appeal decisions or actions pertaining to admissions, programs, evaluation of performance and program retention and completion. Students who decide to file a grievance should follow University student grievance procedures and/or consult with the College Associate Dean for Academic Affairs: Edward M. Olivos at emolivos@uoregon.edu or 541-346-2983.

Classroom Behaviors

All members of the class (both students and instructor) can expect to:

Participate and Contribute: All students are expected to participate by sharing ideas and contributing to the learning environment. This entails preparing, following instructions, and engaging respectfully and thoughtfully with others. While all students should participate, participation is not just talking, and a range of participation activities support learning. Participation might look like speaking aloud in the full class and in small groups as well as submitting questions prior to class or engaging with Discussion posts.

Expect and Respect Diversity: All classes at the University of Oregon welcome and respect diverse experiences, perspectives, and approaches. What is not welcome are behaviors or contributions that undermine, demean, or marginalize others based on race, ethnicity, gender, sex, age, sexual orientation, religion, ability, or socioeconomic status. We will value differences and communicate disagreements with respect. We may establish more specific guidelines and protocols to ensure inclusion and equity for all members of our learning community.

Help Everyone Learn: Part of how we learn together is by learning from one another. To do this effectively, we need to be patient with each other, identify ways we can assist others, and be open-minded to receive help and feedback from others. Don't hesitate to contact me to ask for assistance or offer suggestions that might help us learn better.

Guidelines for using Canvas Discussion:

- Use subject lines that clearly communicate the content of your post
- Write concisely and be aware that humor or sarcasm doesn't always translate in writing.
- Be supportive and considerate when replying to others' posts. This means avoiding use of jargon or inappropriate language, and it means disagreeing with respect and providing clear rationale or evidence to support your different view.
- Keep focused on the topic and reference readings and other class materials to support your points (as applicable).
- Try to use correct spelling and grammar and proofread your submissions. After submitting, use the edit feature to make corrections and resubmit (don't create a new or duplicate post that corrects your error).
- Contribute and interact often!

University course policies

Please read and make sure you are familiar with the important policies that apply to all UO courses, which can be found at [University Course Policies: a Guide for Students](#).

Course Schedule (Check this link for more detailed schedule https://czopluoglu.github.io/educ641_f25/schedule.html)

Week	Date	Topic	Assignments Post Date	Readings
1	09/30 10/2	Introductions, Course Overview Introduction to scientific principles and data analysis		Course Syllabus LSWR, Chapter 1, 2 Light, Singer, Willet, 1990
2	10/7 10/9	Descriptive Statistics for Continuous Data Visualizing Continuous Data	Lab 1 Quiz 1	LSWR, Chapter 5-6
3	10/14 10/16	Introduction to Normal Distribution Sampling distribution Central Limit Theorem	Lab 2 Quiz 2	LSWR, Chapter 9-10
4-5	10/21 10/23 10/28 10/30	One-sample z-test One-sample t-test Independent Samples t-test Dependent samples t-test	Lab 3 Quiz 3	LSWR, Chapter 11, 13 Podcast: Guinness Evans 2020
6	11/4 11/6	Relationship between continuous variables		LSWR, Chapter 15 Gould, Chapter 3
7	11/13	Troubling history of statistics		Clayton 2020
8-9	11/18 11/20 11/25	Relationship between categorical variables	Lab 4 Quiz 4	LSWR, Chapter 12
10	12/2 12/4	Critiques of NHST		
	12/10	Final Assignment is due at midnight		

* The column indicates the dates the homework assignments will be post

APPENDIX A

Dependent samples t-test

Same individuals are measured at two different time points or two different conditions. You compare the same group of individuals under two different time points or two different conditions. Your dependent variable is a continuous outcome (preferably with an interval or ratio scale, but some people also treat ordinal variables with many categories as continuous). Your independent variable is a categorical variable with two levels (categories).

Independent samples t-test

Two independent groups of individuals with a different characteristic (e.g., ELL vs. non-ELL students) are measured only once on the same variable. Or, two independent groups of individuals are measured only once under two different conditions (e.g., the first group is measured under control condition, the second group is measured under the experimental condition). Your dependent variable is a continuous outcome (preferably with an interval or ratio scale, but some people also treat ordinal variables with many categories as continuous). Your independent variable is a categorical variable (ordinal or nominal) with two levels (categories).

Chi-square Test of Independence

Same individuals are measured on two different categorical variables (ordinal or nominal), and you want to see if these two variables are related to each other. Your dependent variable is a categorical variable (ordinal or nominal) with at least two levels (categories). Your independent variable is also a categorical variable (ordinal or nominal) with at least two levels (categories).

Pearson Correlation

Same individuals are measured on two different continuous variables (preferably with interval or ratio scale, but some people also treat ordinal variables with a large number of categories as continuous). You want to see if these two variables are related to each other. Your dependent variable is a continuous variable. Your independent variable is also a continuous variable.

Simple Linear Regression

Same individuals are measured on two different continuous variables (preferably with interval or ratio scale, but some people also treat ordinal variables with a large number of categories as continuous). You want to see if these two variables are related to each other. Your dependent variable is a continuous variable. Your independent variable is also a continuous variable.